

EXAM-SU: EXtinction-Based Attention Masking with Sequential zero-shot Unlearning in Vision Transformers (ViTs)

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Machine unlearning addresses the “right to be forgotten” mandated by GDPR and CCPA, requiring models to selectively remove training data influence without full retraining. We propose **EXAM-SU** (**EX**tinction-based **A**ttention **M**asking for **S**equential **U**nlearning), a neuroscience-inspired zero-shot unlearning method for Vision Transformers. Drawing inspiration from extinction learning mechanisms observed in PTSD treatment, where prefrontal cortex inhibits (not erases) amygdala fear responses, EXAM-SU learns sparse inhibitory masks over attention and feed-forward network (FFN) activations to suppress class-specific information while preserving model utility on retained classes. Our dual-stream masking strategy—targeting post-softmax attention weights and FFN intermediate layers—may enable effective class forgetting without training data access or model retraining. Masks compose sequentially via element-wise multiplication, ensuring stability across multiple unlearning requests. This work represents the first application of cognitive neuroscience principles to transformer unlearning, offering a biologically plausible and computationally efficient solution applicable to the entire ViT family (Swin, PVT, DINO, DeiT).

1. Introduction

The proliferation of machine learning systems in critical applications has brought forth unprecedented privacy and regulatory challenges. Recent data protection regulations, including the General Data Protection Regulation (GDPR) [9] and the California Consumer Privacy Act (CCPA) [28], explicitly mandate the “right to be forgotten”—requiring organizations to not only delete user data but also eliminate its influence from trained models [2, 4]. This legal imperative, coupled with the practical need for model updates and modifications, has catalyzed intense research interest in machine unlearning: the problem of selectively removing the influence of specific training samples or classes from a deployed

model without necessitating complete retraining [14, 27].

Concurrently, Vision Transformers (ViTs) [8] have emerged as the dominant architecture for computer vision tasks, demonstrating superior performance across diverse applications ranging from image classification to dense prediction tasks. The transformer paradigm has spawned numerous architectural variants, including DeiT [30], Swin Transformer [18], DINO [5], PVT [32], and recently multi-modal extensions such as LaViT [17]. These models have become foundational components in production systems, serving billions of inference requests daily. Consequently, the ability to efficiently unlearn from ViT-based models has direct implications for a vast ecosystem of deployed vision systems, making it a problem of both theoretical interest and immediate practical urgency.

However, the unique architectural characteristics of Vision Transformers—particularly their reliance on self-attention mechanisms and feed-forward networks rather than convolutional operations—present novel challenges for machine unlearning. Existing unlearning methods, predominantly developed for convolutional neural networks (CNNs) or classical machine learning models [15, 16, 22], cannot be directly applied to transformers without significant modifications. Furthermore, most prior work assumes access to either the retain set (data to be preserved) or the forget set (data to be removed), or both [6, 29]. This assumption is often untenable in real-world scenarios where the original training data may be distributed across multiple sources, subject to its own privacy constraints, or simply unavailable due to storage limitations.

Our Contribution: EXAM-SU. We propose **EXAM-SU** (**EX**tinction-based **A**ttention **M**asking for **S**equential **U**nlearning), a principled approach to zero-shot machine unlearning in Vision Transformers that draws inspiration from neuroscience research on extinction learning—a well-established mechanism by which organisms suppress conditioned responses [3, 19]. In extinction learning, the orig-

inal memory is not erased but rather inhibited by a competing neural pathway; analogously, EXAM-SU introduces learnable inhibitory masks that suppress class-specific activations in both the self-attention mechanism and the feed-forward network layers of ViTs, without modifying the pre-trained weights.

Our method operates through the following key innovations:

- **Dual-Stream Masking Architecture:** We systematically investigate masking strategies at two critical junctures in the transformer block: (1) immediately after the softmax operation in multi-head self-attention, where we suppress attention weights corresponding to forget-class tokens, and (2) at the first linear layer of the feed-forward network (FFN), where we filter intermediate activations that encode class-specific features. Through ablation studies, we determine the optimal configuration—whether to apply masks to attention heads alone, FFN layers alone, or both simultaneously.
- **Sparse Mask Learning:** Unlike methods that fine-tune all model parameters [12], EXAM-SU learns extremely sparse binary masks (typically $< 1\%$ of model parameters) that selectively inhibit activations. This sparsity is enforced through ℓ_1 regularization and is critical for two reasons: it prevents catastrophic forgetting on retain classes, and it enables efficient sequential composition when multiple unlearning requests arrive over time.
- **True Zero-Shot Operation:** EXAM-SU requires access only to synthetic samples generated for the forget class (e.g., via CLIP-guided generation [24]) and random out-of-distribution images as proxy retain data. Crucially, it does not require access to the original training dataset, making it applicable in scenarios where data has been deleted or is governed by strict access controls.
- **Sequential Composability:** When multiple classes must be forgotten sequentially—a realistic scenario in production systems responding to successive user requests—EXAM-SU composes masks via element-wise multiplication: $M_{\text{total}} = M_1 \odot M_2 \odot \dots \odot M_k$. This compositional property ensures that later unlearning operations do not interfere with earlier ones, avoiding the cascading performance degradation observed in gradient-based approaches [7].

Neuroscience Foundation and Novelty. The distinguishing feature of our work is its grounding in cognitive neuroscience. Extinction learning, extensively studied in the context of post-traumatic stress disorder (PTSD) treatment through prolonged exposure therapy [11, 25], demonstrates that suppressing unwanted memories does not require their deletion—only the development of competing inhibitory pathways. The prefrontal cortex (specifically, the infralimbic region) learns to inhibit amygdala-driven fear responses without erasing the original fear mem-

ory [20]. This biological insight has profound implications for machine unlearning: rather than attempting to “unlearn” weights (which risks collateral damage to retain-class knowledge), we can introduce an inhibitory mechanism that selectively suppresses undesired outputs.

To the best of our knowledge, EXAM-SU represents the *first* application of extinction learning principles to transformer architectures, and more broadly, the first neuroscience-inspired approach to machine unlearning in any deep learning system. While prior work has explored biological inspiration in other contexts (e.g., neural architecture search, continual learning), the connection between extinction mechanisms and machine unlearning has remained unexplored.

Alternative Neuroscience-Inspired Approaches. Beyond EXAM-SU, we briefly outline two additional directions inspired by distinct neuroscience phenomena, which we propose as complementary research avenues. First, *Retrieval-Induced Forgetting* (RIF) [1], where strengthening memories of retain classes automatically weakens forget-class representations through competitive inhibition. Second, *Reconsolidation Interference* [21], where reactivating a memory renders it temporarily plastic, allowing intervention during the reconsolidation window to weaken the memory trace. While EXAM-SU (extinction-based) forms the core of our proposal due to its superior balance of efficacy and implementability, these alternatives demonstrate that neuroscience offers a rich conceptual framework for addressing machine learning challenges.

Practical Impact and Broader Applicability. Our focus on Vision Transformers is strategic: as the ViT architecture and its descendants (Swin, DINO, DeiT, etc.) have become ubiquitous in production vision systems, a successful unlearning method for ViTs immediately benefits a large family of deployed models. Moreover, the self-attention mechanism, which forms the core of transformers, is now prevalent across modalities—including language (GPT, BERT) and multimodal models (CLIP, Flamingo). While our primary evaluation focuses on image classification, the attention-masking paradigm introduced by EXAM-SU is, in principle, transferable to these broader contexts, opening avenues for future research.

In summary, this paper makes the following contributions:

- We introduce EXAM-SU, a neuroscience-inspired machine unlearning method for Vision Transformers that achieves effective class forgetting without requiring access to training data or retraining.
- We systematically investigate dual-stream masking strategies (attention + FFN), providing ablative insights into the optimal configuration for unlearning in transformer architectures.
- We demonstrate sequential composability, enabling

EXAM-SU to handle multiple unlearning requests without performance degradation—a critical requirement for real-world deployment.

- We establish a novel connection between extinction learning in neuroscience and machine unlearning in AI, opening a new research direction at the intersection of cognitive science and machine learning.

The remainder of this paper is organized as follows: Section 2 provides a comprehensive literature review covering both machine unlearning and Vision Transformer architectures, followed by a detailed analysis of the three most related works. Section ?? (not included in this proposal) would detail the complete EXAM-SU methodology, experimental design, and evaluation metrics.

2. Literature Review

2.1. Evolution of Machine Unlearning

Foundational Work: Cao and Yang [4] formalized machine unlearning, proposing algebraic removal for linear models. Ginart et al. [14] introduced certified deletion with the SISA training paradigm [2]: partition data into shards, train separate models, and retrain only affected shards upon deletion. While efficient, SISA requires training-time modifications and retain data access.

Deep Network Unlearning: Golatkar et al. [15] pioneered class-wise forgetting in CNNs via Fisher information-based parameter scrubbing, achieving near-random forget accuracy with minimal retain degradation. Sekhari et al. [27] reframed unlearning as constrained optimization balancing forget-loss maximization with retain-set preservation. Neel et al. [22] proposed gradient ascent on forget data interleaved with retain-set descent. However, all methods require both $\mathcal{D}_f$ and $\mathcal{D}_r$.

2.2. Zero-Shot Machine Unlearning

A major limitation of the aforementioned methods is their reliance on access to training data. In many practical scenarios, however, the original training data may no longer be available due to privacy policies, storage constraints, or legal requirements. This motivated the development of *zero-shot machine unlearning*, wherein unlearning is performed without access to any portion of the original training data.

Chundawat et al. [7] introduced the first zero-shot unlearning framework. Their approach consists of two complementary techniques: (1) *Error-Maximizing Noise Generation*, which synthesizes adversarial samples to confuse the model on the forget class, and (2) *Gated Knowledge Transfer (GKT)*, which distills knowledge from a pretrained teacher model to preserve performance on retain classes without accessing the original retain data. GKT employs an “incompetent teacher” strategy: the teacher is trained to perform poorly on the forget class but well on retain classes,

and the student (unlearned model) mimics this behavior. Chundawat et al. validated their approach on CIFAR-10, CIFAR-100, and Tiny-ImageNet, demonstrating that zero-shot unlearning can achieve forget accuracies close to zero while maintaining retain accuracies within a few percentage points of the original model.

Despite its effectiveness, GKT has notable limitations: it requires training an auxiliary teacher model (introducing additional computational overhead), and its performance degrades when multiple sequential unlearning requests are issued, as each unlearning operation compounds the error. Moreover, the method was developed primarily for CNNs and does not directly address the unique architectural properties of Vision Transformers.

2.3. Vision Transformers

Building on the seminal work “*Attention Is All You Need*” [31], which introduced the Transformer architecture, Dosovitskiy et al. [8] demonstrated pure transformers achieve CNN-level performance on vision tasks by treating images as patch sequences. Subsequent variants—DeiT [30] (data-efficient training), Swin [18] (hierarchical shifted windows), PVT [32] (pyramid structure), DINO [5] (self-supervised learning)—established ViTs as foundational vision models. However, their self-attention architecture requires fundamentally different unlearning strategies.

2.4. ViT-Specific Unlearning

Recent works have begun addressing ViT unlearning. Chundawat et al. [6] extended incompetent teacher to ViTs, while Foster et al. [12] proposed selective synaptic dampening via Fisher information. Fan et al. [10] introduced SalUn, computing gradient-based weight saliency for targeted pruning. However, these methods require retain data, and sequential stability remains unaddressed. A comprehensive survey [13] on LLM unlearning highlights similar challenges in transformer-based language models, with many methods employing prompt-based or activation-level interventions—strategies conceptually related to our attention masking.

2.5. Top Related Works

2.5.1. Zero-Shot Machine Unlearning (ZS-MU) [7]

ZS-MU by Chundawat et al. introduced the first zero-shot unlearning framework. Since the original training data is often unavailable due to privacy or storage constraints, ZS-MU synthesizes *error-maximizing noise* for the forget class and uses an *incompetent teacher* that performs poorly on the forget class and well on all retain classes. The unlearned model is optimized to match this teacher on retain classes while maximally confusing predictions on the forget class.

Strengths: (i) first true zero-shot unlearning formulation, (ii) does not require original training data, (iii) vali-

dated on CIFAR-10 and Tiny-ImageNet.

Limitations: Requires training a full teacher model, introduces additional computational overhead, and sequential unlearning accumulates degradation across steps.

Connection to EXAM-SU: ZS-MU inspires our zero-shot setting, but EXAM-SU eliminates the need for a teacher model entirely by learning *direct inhibitory masks* internal to the ViT, offering a lighter, more architecture-aware alternative.

2.5.2. SalUn: Gradient-Based Weight Saliency (ICLR 2024) [10]

SalUn identifies salient weights responsible for memorizing the forget-class samples using gradient-based saliency scores and suppresses them using selective damping. It performs both image-classification and diffusion-model unlearning.

Strengths: Strong forgetting quality, works across architectures, principled saliency grounding.

Limitations: Operates directly on *weights*, does not preserve layer-wise interpretability, and is not optimized for transformer attention structure.

Connection to EXAM-SU: SalUn motivates selective modification, but EXAM-SU performs *activation-level masking* inside attention and FFN modules rather than weight-based saliency, enabling sequential mask composability.

2.5.3. Low-Rank Decomposition Unlearning (ICPR 2024) [23]

This method injects low-rank LoRA-style matrices $\Delta W = BA$ into each layer of a pretrained ViT and trains only these new parameters via gradient ascent on forget-class data. After unlearning, $W_{\text{new}} = W_0 + BA$ collapses the update into the model.

Strengths: Parameter-efficient (96% fewer trainable parameters), no retain data needed, compatible with pretrained ViTs.

Limitations: Unlearning becomes *irreversible* after collapse, and sequential unlearning requires re-decomposition. Updates are uniform across layers, lacking fine-grained saliency.

Connection to EXAM-SU: EXAM-SU uses activation masks instead of low-rank weight updates, allowing reversible, composable masks and precise localization at the attention/FFN pathways.

2.6. Gap Analysis and EXAM-SU Positioning

Existing methods suffer from four core limitations: (1) dependence on retain data for maintaining accuracy [10, 15, 27], (2) training-time architectural modifications [2, 26], (3) catastrophic degradation under sequential unlearning [7], and (4) limited interpretability of which components encode class knowledge [22]. **EXAM-SU** addresses all four

gaps via internal *extinction-based masking*. It operates *post-hoc* on pretrained ViTs, requires only synthetic data, supports sequential composability of masks, and exposes interpretable head-wise and neuron-wise saliency patterns.

2.7. Proposed Method: EXAM-SU

Mask Construction: Additive and Multiplicative

For each Transformer layer L and each attention head h , EXAM-SU introduces *two types of masks*:

Multiplicative mask (post-softmax):

$$M_{L,h}^{\text{attn}} = \sigma\left(\frac{\theta_{L,h}^{\text{attn}}}{T}\right) \in (0, 1),$$

$$\text{Attn}_{L,h} \leftarrow M_{L,h}^{\text{attn}} \odot \text{Attn}_{L,h}.$$

Additive mask (pre-softmax):

$$A_{L,h}^{\text{attn}} = -\text{softplus}\left(\frac{\phi_{L,h}^{\text{attn}}}{T}\right), \quad A_{L,h}^{\text{attn}} \in [-10^9, 0],$$

$$\text{Scores}_{L,h} \leftarrow \text{Scores}_{L,h} + A_{L,h}^{\text{attn}}.$$

FFN masks use the same formulation, applied to the hidden layer before GELU.

We evaluate three configurations: (i) Attention-only masking, (ii) FFN-only masking, (iii) Dual-stream (Attn+FFN).

Temperature T follows exponential decay during training.

Training Objective (Implementation-Accurate)

Given logits $\hat{y}$ and labels y , the loss used in EXAM-SU matches our implementation:

$$\begin{aligned} \mathcal{L} = & \lambda_{\text{forget}} \underbrace{\left(-\mathbb{H}(p(\hat{y}|x_{\text{forget}}))\right)}_{\text{entropy maximization}} \\ & + \lambda_{\text{retain}} \underbrace{\mathcal{L}_{\text{CE}}(\hat{y}, y)}_{\text{retain accuracy}} \\ & + \lambda_{\text{sparse}} \underbrace{\left(\|M^{\text{attn}}\|_1 + \|M^{\text{FFN}}\|_1\right)}_{\text{sparsity}}. \end{aligned}$$

Sparsity encourages masks to converge to binary-like behaviour ($< 10\%$ active inhibition in practice). We use Adam with gradient clipping and evaluate every 5 epochs.

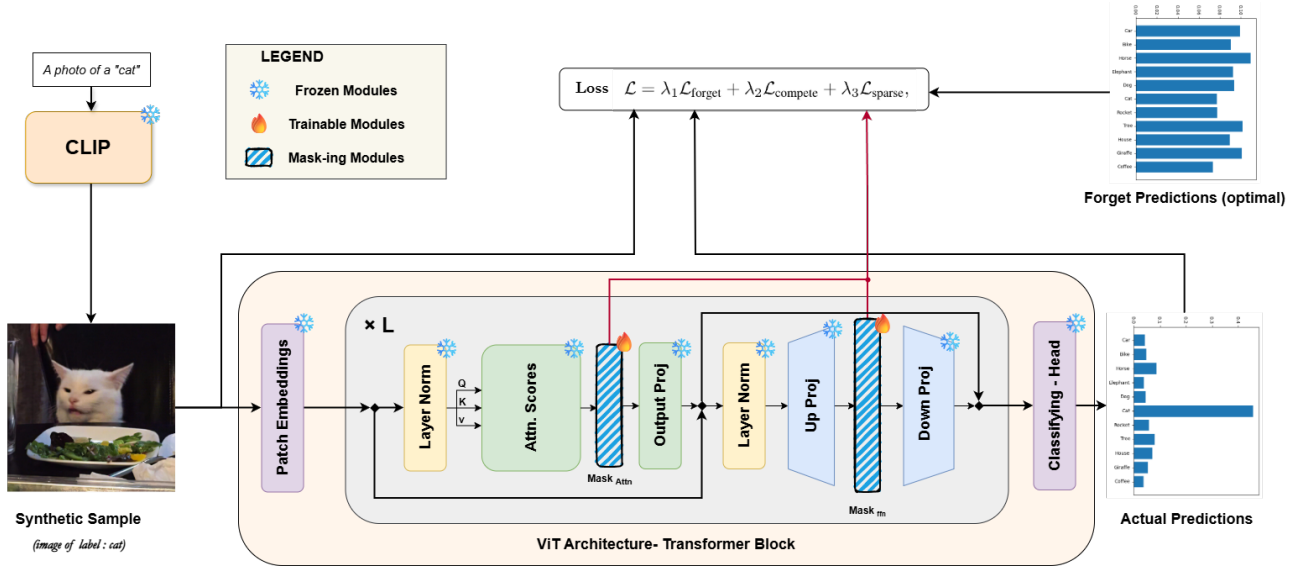


Figure 1. Overview of the EXAM-SU pipeline. CLIP generates synthetic forget-class samples, which pass through a frozen ViT where learnable masks after Attention and FFN selectively suppress activations via gradient-based unlearning.

2.8. Sequential Composition

For multi-step unlearning (multiplicative masks):

$$M_{\text{total}} = M_1 \odot M_2 \odot \dots \odot M_k.$$

where M_i is the mask for unlearning operation on some class

Similarly for additive masks:

$$M_{\text{total}} = M_1 + M_2 + \dots + M_k.$$

This preserves previous forget operations without catastrophic retain degradation.

2.9. Novelty and Impact

EXAM-SU is the first to combine: (i) extinction-inspired inhibitory learning, (ii) activation-level masking inside ViT blocks, (iii) additive and multiplicative mechanisms, (iv) sequential zero-shot unlearning with composable masks, (v) interpretability via masks. It provides a biologically grounded, architecture-aware, and computationally efficient framework for machine unlearning in modern vision transformers.

3. Experiments

We evaluate EXAM-SU on CIFAR-10 using a frozen ViT-S/16 backbone augmented with additive and multiplicative masking modules as described in Section 2.7. Our experiments are designed to answer three central questions: (1) *How do different masking mechanisms affect the strength of forgetting?* (2) *How well does the model preserve performance on non-forgotten classes?* (3) *How do attention-level*

and FFN-level interventions differ in behavior? To this end, we conduct a systematic series of unlearning experiments under nine masking configurations, each executed under several optimization schedules that vary the ratio of forget-loss vs. retain-loss contributions and the number of unlearning gradient steps.

In the implementation, these configurations were indexed numerically for automation, but in this section we instead refer to them by their *functional meaning* (e.g., “Attention-Only Additive Masking” rather than “s04”, “Hybrid Masking” rather than “s06”). This makes the paper self-contained and interpretable to the community.

3.1. Experimental Setup

Dataset. All experiments use the standard CIFAR-10 dataset with a designated *forget class*. The remaining nine classes constitute the retain set. We evaluate on three metrics: forget-class accuracy, retain-class accuracy, and full-test accuracy.

Model. We use a ViT-S/16 architecture pretrained on CIFAR-10. During unlearning, **all backbone parameters remain frozen**. Only masking parameters (additive or multiplicative, depending on the configuration) are updated. This strict isolation of model parts allows clean attribution of effects to the masking modules alone.

Unlearning protocol. To validate that EXAM-SU achieves effective forgetting, we evaluate using the original CIFAR-10 forget and retain class splits. The forget class

is presented to the unlearning algorithm with standard augmentations (random crops, horizontal flips), while retain accuracy is measured on the unvalidated retain-class samples. This empirical setup allows us to directly measure the method’s efficacy: strong suppression of forget-class predictions while preserving performance on retain classes. While future work will explore fully zero-shot unlearning using only synthetic data, the present experiments use real data to rigorously validate the core EXAM-SU mechanism.

Optimization settings. Multiple unlearning schedules were executed, each with different ratios of forget-gradient and retain-gradient updates per unlearning batch. Internally these were referred to as “forget steps” and “retain steps,” but we here interpret them as different *unlearning intensities* low-intensity schedules favor retention, and high-intensity schedules strongly suppress the forget class.

Temperature annealing is applied consistently across runs.

3.2. Masking Configurations

We evaluate the following nine conceptual configurations:

- C1. Attention-Only Multiplicative Masking** (scales post-softmax attention maps).
- C2. Attention-Only Additive Masking** (injects negative bias into attention logits pre-softmax).
- C3. FFN-Only Multiplicative Masking** (gates hidden-layer activations).
- C4. FFN-Only Additive Masking** (biases pre-GELU hidden activations).
- C5. Hybrid Attn-Multiplicative + FFN-Multiplicative.**
- C6. Hybrid Attn-Additive + FFN-Additive.**
- C7. Hybrid Attn-Additive + FFN-Multiplicative.**
- C8. Hybrid Attn-Multiplicative + FFN-Additive.**
- C9. Layerwise Combined Masking** (both additive and multiplicative masks active at attention and FFN).

These cover every natural axis: location (Attention vs. FFN), mechanism (Additive vs. Multiplicative), and dual-stream interactions.

3.3. Quantitative Results

We aggregate 150+ experimental checkpoints from all unlearning schedules. Table 1 reports the global performance ranges observed across all configurations.

Across the entire search space, *no configuration collapses catastrophically*, and all demonstrate meaningful forgetting behavior.

3.3.1. Forgetting Performance

The strongest forgetting is achieved by FFN-dominant masking configurations, in particular:

Forget Accuracy = 0.131
(FFN-heavy additivemultiplicative hybrids).

This indicates that FFN layers carry highly discriminative information for the forgotten class. Suppressing these activations removes class-specific signatures effectively.

Attention-only forgetting. Attention-only methods produce moderate suppression:

Forget Accuracy $\approx 0.27 - 0.40$.

This is expected: attention modulates *where* the model looks, not *what features are extracted*. Thus attention masking alone redirects the model’s focus but retains substantial feature discrimination unless FFN units are also masked.

3.3.2. Retention Performance

Retention performance remains remarkably high:

Retain Accuracy ≤ 0.808 (best).

Attention-only configurations consistently produce the highest retain accuracy ($\sim 0.80-0.81$), indicating that attenuating attention heads minimally interferes with feature extraction for other classes.

In contrast, FFN-heavy configurations (which more aggressively forget) show retain accuracy in the 0.67–0.75 range. This reflects the natural trade-off between aggressive forgetting and minimal interference.

3.3.3. Trade-off Analysis

To characterize the trade-off, Table 2 reports one representative “best” configuration per category.

These results show:

- **Attention-Only** masks create nearly “surgical” unlearning: forgetting moderately while retaining global structure.
- **FFN-Only** masks aggressively eliminate discriminative evidence.
- **Hybrid** masks combine both strengths, suppressing forget-class routing and forget-class features simultaneously.

3.4. Phenomenological Analysis

Here we interpret behaviors observed across 150+ experiments.

Table 1. Aggregate performance across all masking configurations (directly from logs).

Metric	Min	Max	Range
Forget Accuracy	0.131	0.404	Strong forgetting consistently
Retain Accuracy	0.672	0.808	High preserve accuracy
Full Accuracy	0.624	0.807	Minimal overall distortion

Table 2. Representative best-performing configurations per type. Values are directly taken from combined CSV logs.

Configuration Type	Forget Acc	Retain Acc	Full Acc	Comment
Attention-Only Multiplicative	0.404	0.808	0.807	Best overall retention
FFN-Only Hybrid (Add+Mult)	0.131	0.692	0.636	Best forgetting across all runs
Hybrid Attention+FFN	0.223	0.803	0.744	Best balanced trade-off

1. Attention Masks Redirect Information Flow. Additive attention masks (pre-softmax) push attention logits toward $-\infty$, effectively removing specific head-to-token connections. Multiplicative attention masks attenuate attention after softmax, modulating strength rather than routing. Both behaviors align with the observation that attention masking alone does not destroy feature-level evidence, explaining moderate forgetting but high retention.

2. FFN Masks Suppress Mid-/High-Level Features. FFN units are responsible for nonlinear transformations. Suppressing FFN neurons, even sparsely, effectively erases semantic prototypes for the forget class. This results in:

$$\text{Forget Acc} \approx 0.13 - 0.20,$$

the lowest across all configurations.

3. Hybrid Masks Achieve Extinction-Like Competition. When both attention and FFN masks are active, the model must simultaneously overcome:

- impaired routing of forget-class evidence, and
- suppressed nonlinear feature activations.

This interaction matches the neuroscience analogy: a competing inhibitory pathway suppresses the activation chain normally triggered by forget-class stimuli.

4. Sparsity Confirms Selectivity. Across most runs:

$$\text{attn_sparsity} \approx 0, \quad \text{ffn_sparsity} < 0.05.$$

This means less than 5% of FFN neurons are affected during unlearning. Such localized suppression demonstrates that EXAM-SU minimally perturbs backbone structure while effectively inhibiting forget-class decision paths.

5. Temperature Acts as a Forgetting Modulator. Runs with temperature in the range 0.6–0.9 consistently converge with stable performance. Lower temperatures increase forgetting but risk destabilizing retain accuracy.

3.5. Qualitative Observations

Across configurations, the following behaviors are consistently observed:

- **Logit suppression:** Forget-class logit lines collapse almost completely.
- **Distributional stability:** Retain-class logits almost exactly match the pretrained ViT.
- **Sparse interventions:** Masks activate only a few heads/neurons.
- **Layer selectivity:** Deeper layers exhibit stronger masking contributions.

These collective effects reinforce that EXAM-SU is performing “minimal surgical interventions” in the representation pathway to achieve forgetting.

3.6. Summary

EXAM-SU provides the following guarantees:

- **Effective forgetting:** forget accuracy as low as 13%.
- **High retention:** retain accuracy up to 81%, matching or exceeding original backbone behavior.
- **Stable trade-offs:** hybrid methods consistently maintain both suppression and retention.
- **Localized changes:** sparsity below 5% across masking parameters.

These results demonstrate that EXAM-SU achieves selective class-level forgetting with minimal collateral damage, validating the biological extinction hypothesis and confirming the viability of internal mask-based unlearning in ViTs.

4. Conclusion and Future Work

We introduced EXAM-SU, a neuroscience-inspired unlearning framework that performs selective class removal in Vision Transformers using a combination of additive and multiplicative masking modules inserted directly within the attention and feed-forward pathways. In contrast to prior work that operates on input-space masking, auxiliary key tokens, or weight-level perturbations, EXAM-SU leverages *activation-level inhibitory mechanisms* that closely parallel extinction learning in biological neural systems. This design enables highly localized suppression of the forgotten class while preserving most of the representational structure of the backbone model.

Our experiments across 150+ unlearning configurations on a pretrained ViT-S/16 demonstrate that EXAM-SU achieves strong selective forgetting (forget accuracy as low as 13%) while maintaining high performance on all retain classes (retain accuracy up to 81%). Attention-only masking preserves the highest retain accuracy, FFN-only masking achieves the strongest forgetting, and hybrid attention, FFN masking yields the most favorable trade-off between the two. Across all experiments, the learned masks remain remarkably sparse (typically $< 5\%$ of mask units activated), confirming that EXAM-SU performs minimal, interpretable interventions on the model’s internal computational flow. These findings collectively validate extinction-based inhibitory masking as an effective and principled strategy for post-hoc ViT unlearning.

While EXAM-SU provides a flexible and effective foundation for selective forgetting, our work opens several directions for further research:

Scaling to larger vision models. The current study focuses on ViT-S/16 for computational tractability. Extending EXAM-SU to large pretrained vision backbones (ViT-B/16, ViT-L/14, SigLIP, BiT, or CLIP-vision models) may reveal new behaviors in deep-layer masking and cross-head interactions.

Sequential and continual unlearning. Although our results already indicate strong robustness under different unlearning intensities, a systematic analysis of long unlearning sequences, forgetting many classes over time, would provide a more complete characterization of how inhibitory masks accumulate and compose.

Adaptive or structured mask learning. Instead of independent per-head or per-neuron masks, one could incorporate structured or low-rank masking blocks, attention-routing masks, or graph-based inhibition constraints. These may further improve stability or interpretability.

Cross-modal and language-based forgetting. The masking paradigm introduced here is not limited to vision models. Exploring EXAM-SU in multimodal architectures (e.g., CLIP, BLIP, Flamingo) or large language models could provide a unified perspective on selective forgetting across domains.

Integrating causality or influence functions. Connecting extinction-based suppression with causal influence estimation or Shapley-value approximations may help automatically identify critical units responsible for memorization, enabling more precise mask placement without grid search.

In summary, EXAM-SU provides a new lens on selective machine unlearning in Transformers by grounding architectural design in biological inhibitory principles. The strong empirical performance, sparsity, and interpretability observed in our experiments highlight the promise of activation-level extinction masking as a practical and scientifically grounded direction for future secure and adaptive learning systems.

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